PGLink
Distance-Based PG Accommodation Discovery System

Final Year Project Report

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1. Introduction

PGLink is a web-based software application designed to help students find nearby PG (Paying Guest) accommodations based on their college location. The platform simplifies the process of searching for suitable housing by providing distance-based results.

2. Problem Statement

Students relocating to new cities often face difficulty in finding affordable and nearby PG accommodations. Existing platforms do not prioritize distance effectively and may provide unorganized results, making the search process time-consuming.

3. Proposed Solution

PGLink allows users to search PG accommodations by entering a college name. The system identifies the college from the database and displays nearby PG listings sorted by distance, helping users make better decisions quickly.

4. Working of the System

College Selection: The user selects a college, which triggers a backend search for its specific GPS coordinates (Latitude/Longitude).

Proximity Search: The system uses a **Spatial Query (ST_DistanceSphere)** to calculate the physical distance between the college and all listed PGs.

Distance Sorting: Results are dynamically sorted in ascending order (closest first) before being sent to the React frontend.

SMS Integration: Owners can register PGs by interacting with a conversational SMS bot that saves data to the central database in real-time.

5. Key Features

- Distance-based PG listing
- College-based search system
- Handling of multiple college matches
- SMS-based PG listing support

6. Technology Stack

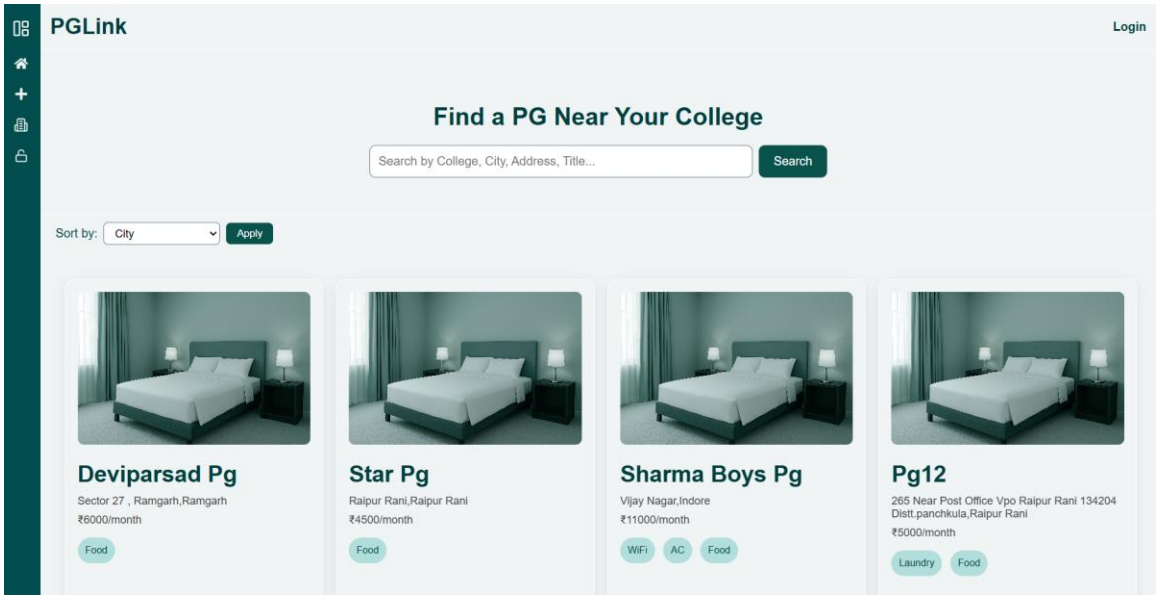
Frontend: React.js, CSS

Backend: Node.js, Express.js

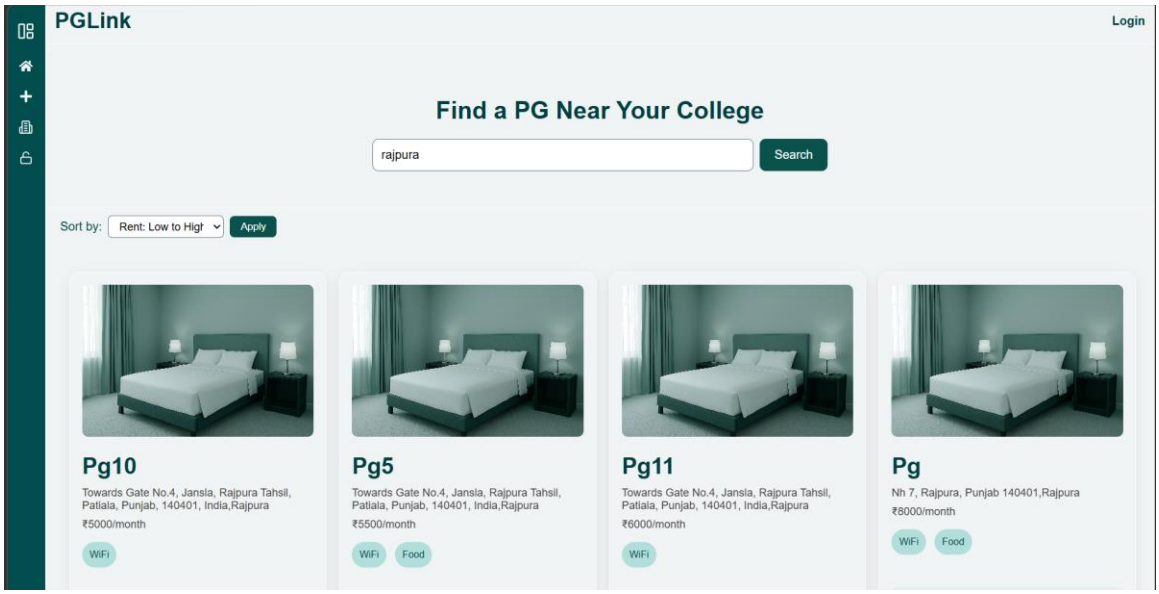
Database: MySQL

Languages: JavaScript, HTML, CSS

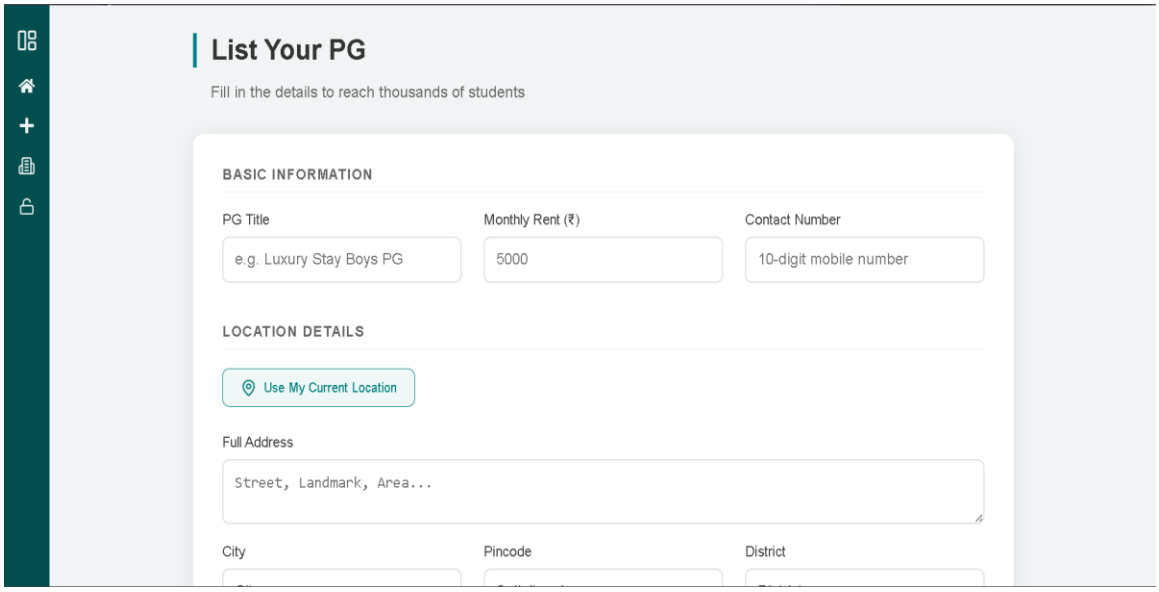
- Home Page



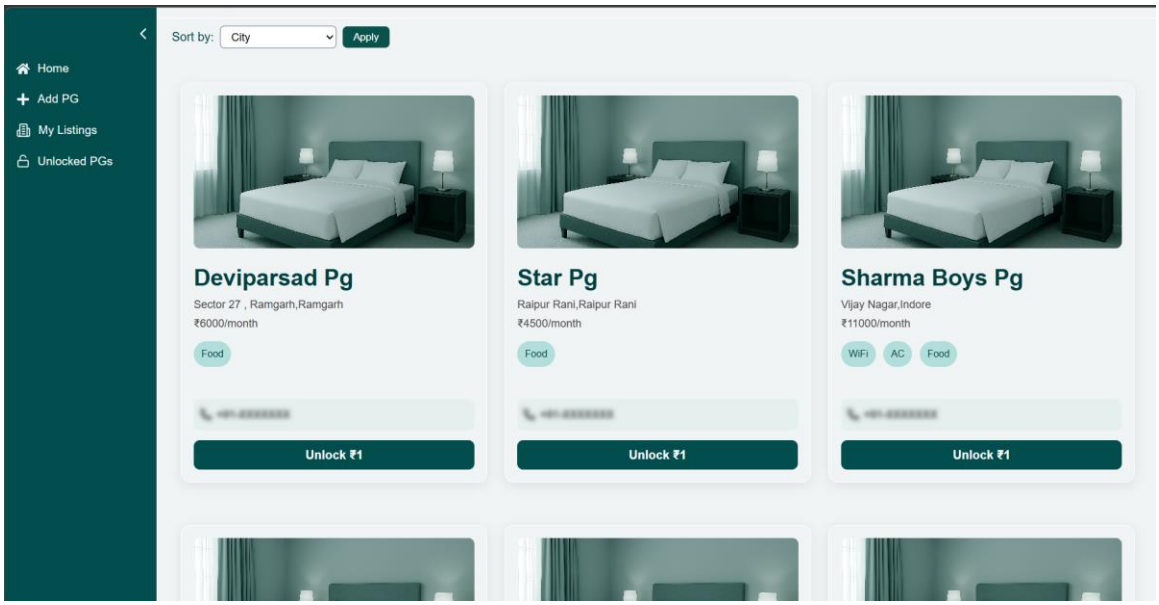
• Search Results Page



• Add PG Form



• Dashboard



8. Conclusion

PGLink provides a simple and effective solution for students to find nearby PG accommodations. By focusing on distance-based results, it improves the search experience and saves time.

9. Declaration

We hereby declare that the project titled 'PGLink: Distance-Based PG Accommodation Discovery System' is an original work carried out by us under the guidance of our mentor. This project has not been submitted to any other university or institution for the award of any degree or diploma.

The work presented in this report is a result of our own efforts and all sources of information have been duly acknowledged.